

TRAP whole-brain analysis — colleague explanation

Figure + meaning + code for every step used in the final story

Findings: $A = \text{ORBm} / \text{BMAp} (\text{Active} \gg \text{Passive}) \cdot B = \text{COAa} (\text{Passive-biased})$

Code: https://github.com/limserenahansol/TRAP_analysis

Built for AI-benchmark navigation (final-PDF figures only)

1 · Biological aim

- Primary: find regions with the largest, most consistent Active >> Passive TRAP density across morphine phases.
- Secondary: find Passive-biased regions (Passive > Active at Post + Withdrawal) that may track generalized motivation.
- Design: Active (contingent morphine SA) vs Passive (yoked) · During → Post → Withdrawal → Reinstatement.
- Readout: TRAP+ cell density (cells / sample volume mm³), L/R pooled · Allen atlas · calculated_mm3.
- Why it matters: Active-selective + Passive-flat → opioid-related seeking; Passive-biased → patient-relevant / general seeking candidate.

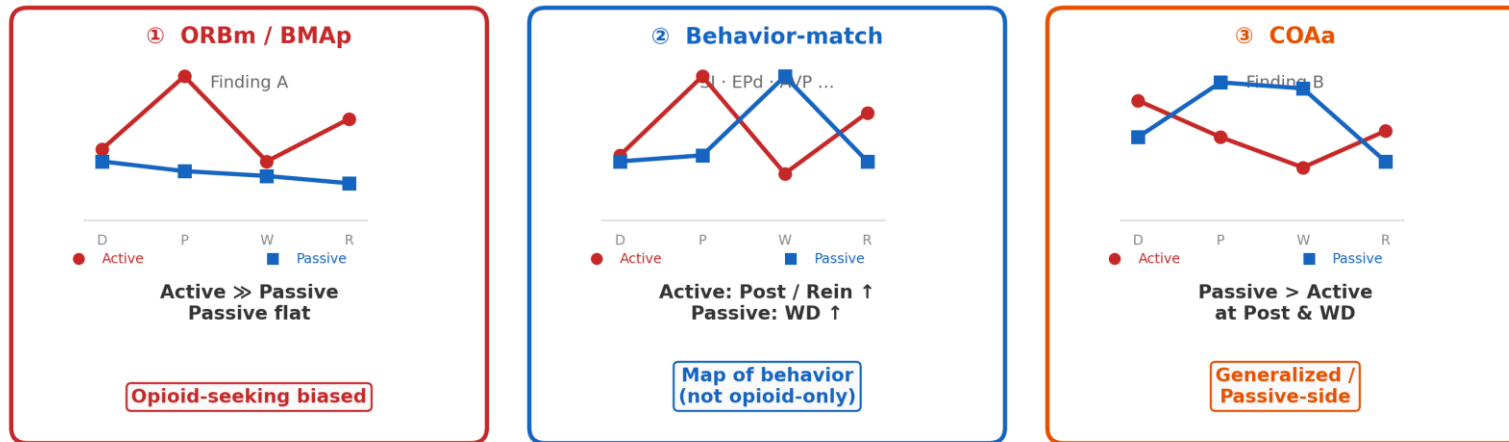
2 · Two findings (final story)

- FINDING A — ORBm & BMAp: Active >> Passive; Passive flat → dual-role craving (opioid seeking + WD negative affect).
- FINDING B — COAa: Passive > Active at Post + Withdrawal → cue/context + WD state → generalized-motivation candidate.
- Why keep both: A isolates opioid-related Active seeking; B is the Passive / patient-relevant contrast.
- Why not only behavior-matched regions for A: those track both opioid craving and Passive generalized motivation.

3 · Framing — opioid seeking vs general seeking

Three circuit types

Active = opioid + seeking · Passive = same drug, no seeking



- WHAT: schematic separating Active-selective craving from Passive / general seeking.
- WHY: Passive-flat at ORBm/BMAP argues against pure general-seeking accounts.
- COAa is the Passive-side / generalized candidate to manipulate separately.
- AI check: agent should state both stories, not only Active>>Passive.

4 · Workflow funnel (why cluster first, then zoom in)

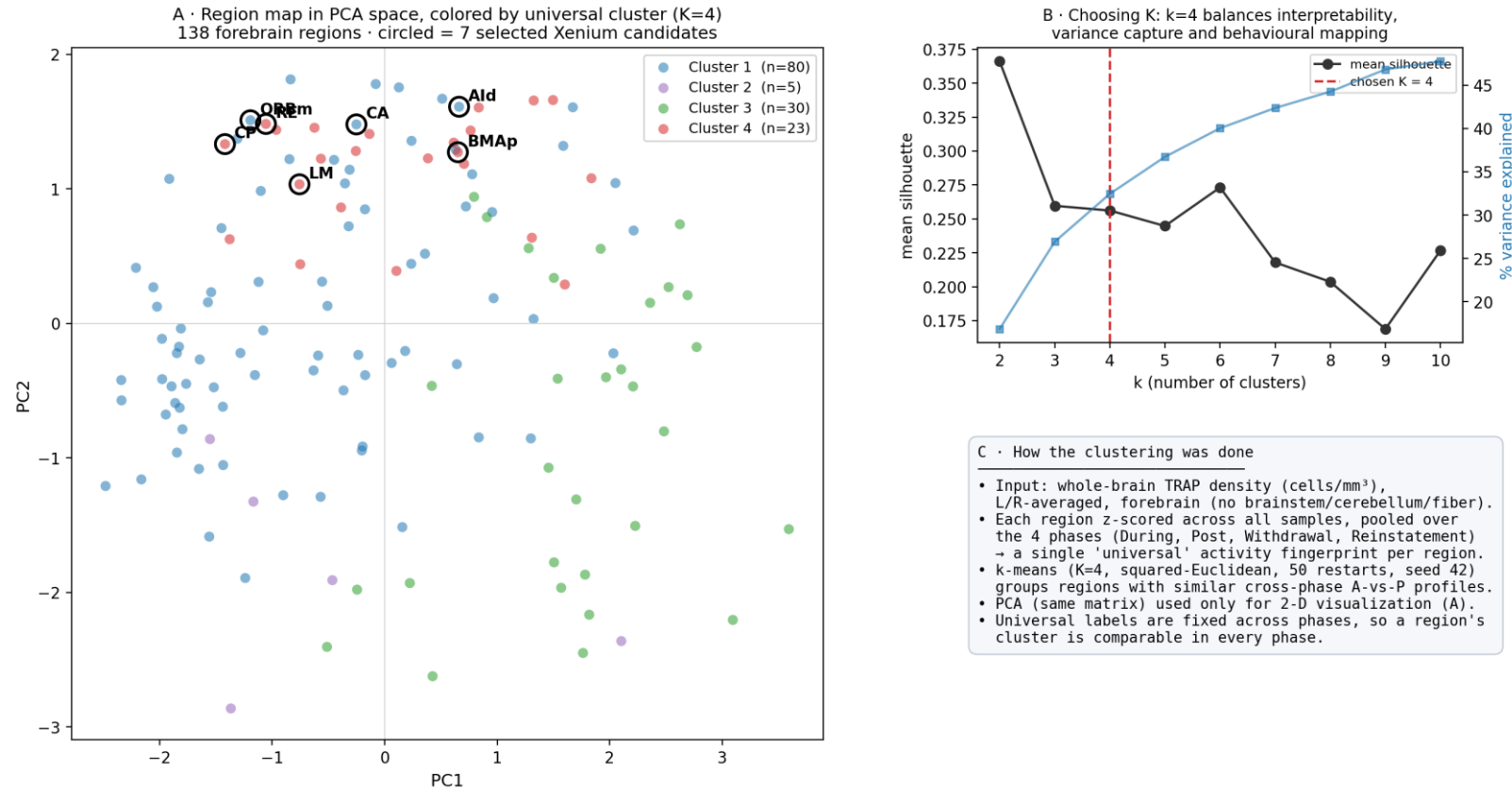
~800+ atlas → hierarchy567 → forebrain_no_bs → K=4 → clusters 1&4 → 7 → ORBm/BMAp + COAa

- Problem: hundreds of atlas regions → testing one-by-one is under-powered and unstructured.
- Step 1 FILTER: forebrain_no_bs (~138) · universal PCA display + k-means K=4 on Active–Passive fingerprint across 4 phases.
- Step 2 KEEP: only clusters 1 & 4 match motivation timeline (A>P at Post & Reinstatement).
- Step 3 SHORTLIST: cluster4 BMAp/LM/RE/CP + cluster1 ORBm/CA/Ald (= 7 regions).
- Step 4 ZOOM IN: density + cell count + mouse-level → only ORBm & BMAp survive as Finding A; Finding B from cluster 3 → COAa.

MATLAB: RUN_PIPELINE_ALL.m · Step_03_region_clustering... · Step_13_universal_cluster_viz/...
Python story figures: trap_region_selection_ppt.py · trap_region_zoomin.py · trap_build_pptx.py

5 · Fig 1 — Clustering method + PCA map

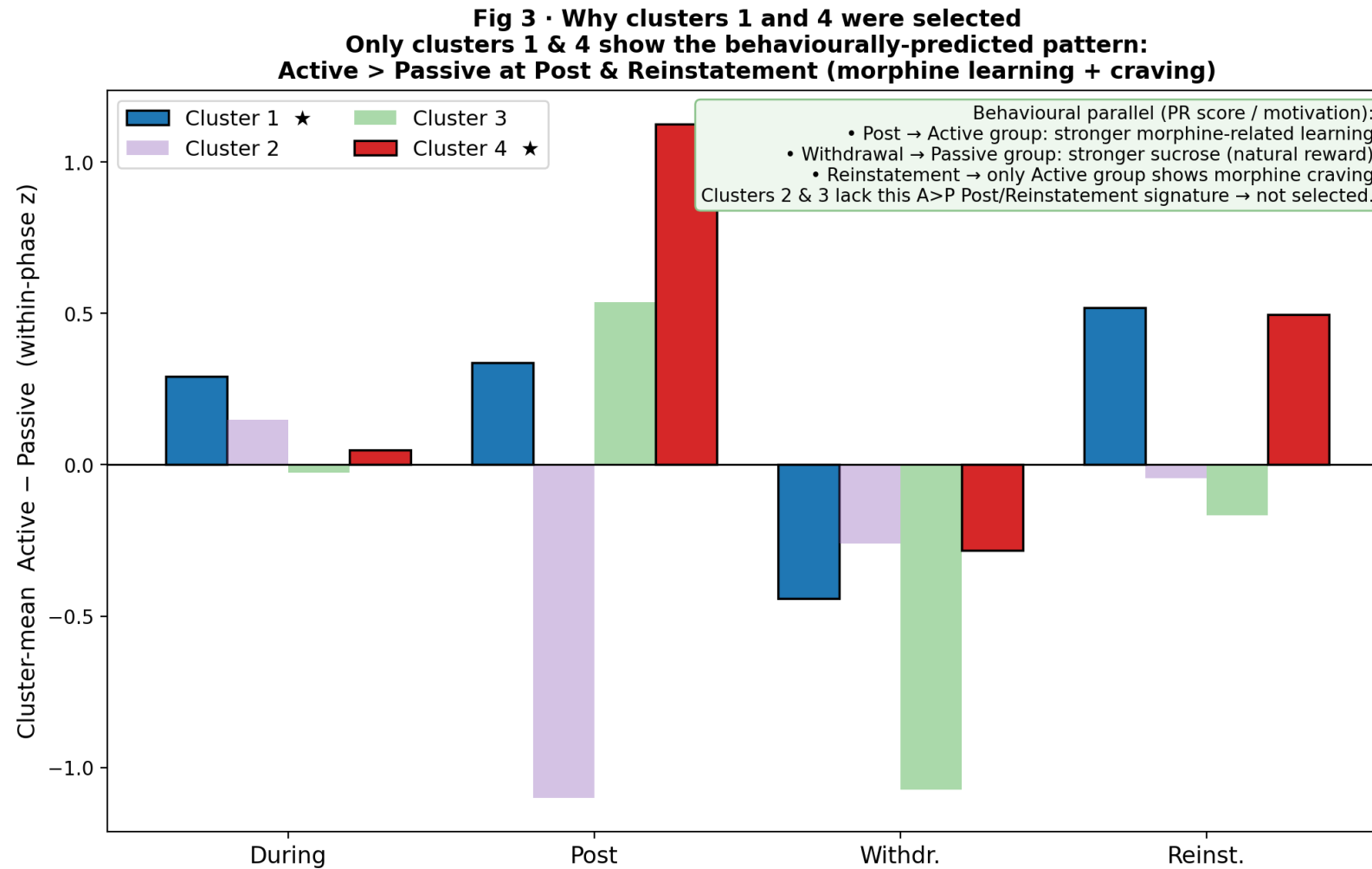
Fig 1 · Universal PCA + k-means clustering of whole-brain TRAP activity



- WHAT: 138 forebrain regions in PC1–PC2, colored by K=4; selected 7 circled.
- Method: z-score region profile across samples → k-means (sq-Euclidean, 50 restarts).
- PCA is 2-D display only; clusters use the full profile.
- AI check: keep fixed Step-3 K=4 labels; silhouette/elbow are supporting only.

CODE: `trap_region_selection_ppt.py` → `region_selection_PPT/Fig01_...`
Upstream MATLAB: Step_03 + Step_13 (`roster 02_cluster_region-roster.csv`)

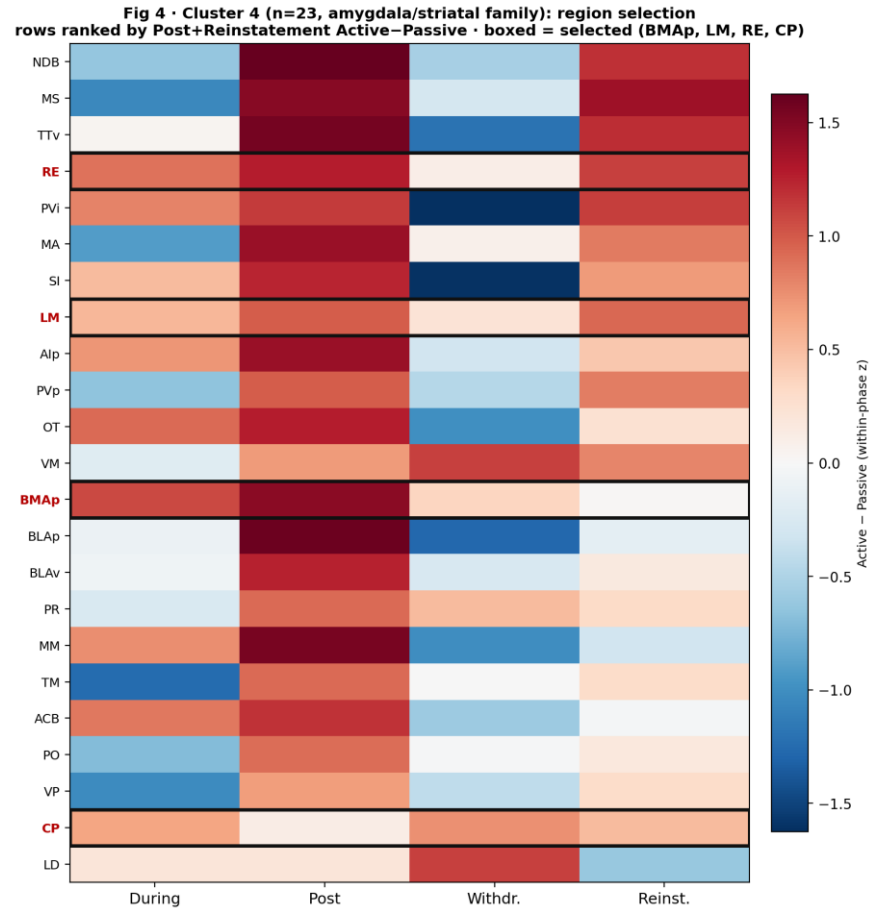
6 · Fig 3 — Why clusters 1 and 4



- WHAT: per-phase bars of cluster-mean Active - Passive.
- Only clusters 1 & 4 show A>P at Post & Reinstatement (motivation timeline).
- Post ≈ Active morphine learning; WD ≈ Passive sucrose shift; Rein ≈ Active craving.
- AI check: select clusters by phase pattern, not silhouette alone.

CODE: `trap_region_selection_ppt.py` → `Fig03_why_clusters_1_and_4.png`
Inputs: `cluster_AP_split/Cluster{N}_region_AP_direction.csv`

7 · Fig 4 — Cluster 4 shortlist (BMAp, LM, RE, CP)

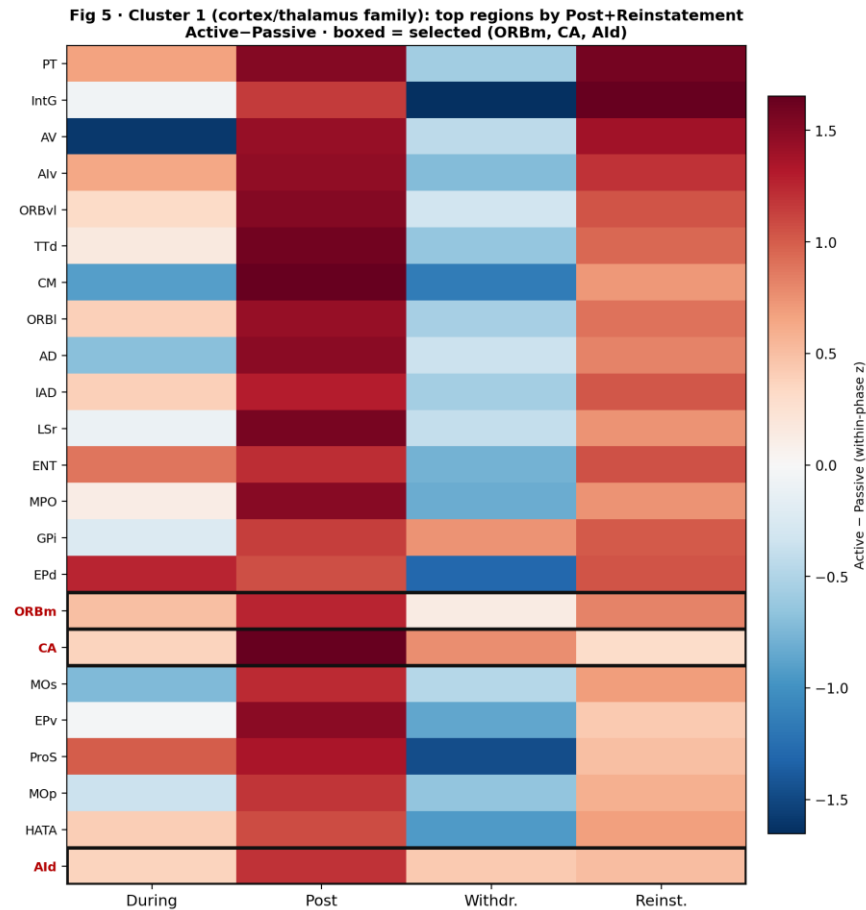


Selection rule (3 criteria, not score alone): (1) member of behaviourally-aligned cluster 4; (2) Active > Passive in the craving-relevant phases (Post/Reinstatement); (3) canonical addiction-circuit node with a defined manipulation target. Other statistically-elevated cluster-4 regions (e.g. NDB, MS, TTv) were deprioritised as less specific / harder to target. BMAp, LM, RE, CP satisfy all three and agree with the independent top-N ranking.

- WHAT: cluster-4 regions × phase heatmap of Active – Passive.
- Boxed / selected: BMAp, LM, RE, CP (A>P in craving-relevant phases).
- BMAp later survives zoom-in as Finding A (with ORBm).
- AI check: recover BMAp from cluster-4 AP shortlist.

CODE: trap_region_selection_ppt.py → Fig04_cluster4_region_selection*.png
MATLAB support: shared/trap_cluster_split_by_AP_direction.m (Step 13)

8 · Fig 5 — Cluster 1 shortlist (ORBm, CA, Ald)



Selection rule (3 criteria, not score alone): (1) member of behaviourally-aligned cluster 1; (2) Active > Passive at Post/Reinstatement; (3) canonical addiction-relevant cortical/limbic node. Higher-scoring but less interpretable nodes (e.g. PT, IntG, AV thalamic nuclei) were deprioritised. ORBm, CA, Ald satisfy all three and agree with the independent top-N ranking.

- WHAT: cluster-1 regions ranked / heatmapped by Active – Passive.
- Selected shortlist: ORBm, CA, Ald.
- ORBm later survives zoom-in as Finding A (with BMap).
- AI check: recover ORBm from cluster-1 shortlist.

CODE: `trap_region_selection_ppt.py` → `Fig05_cluster1_region_selection*.png`
MATLAB support: `shared/trap_cluster_split_by_AP_direction.m` (Step 13)

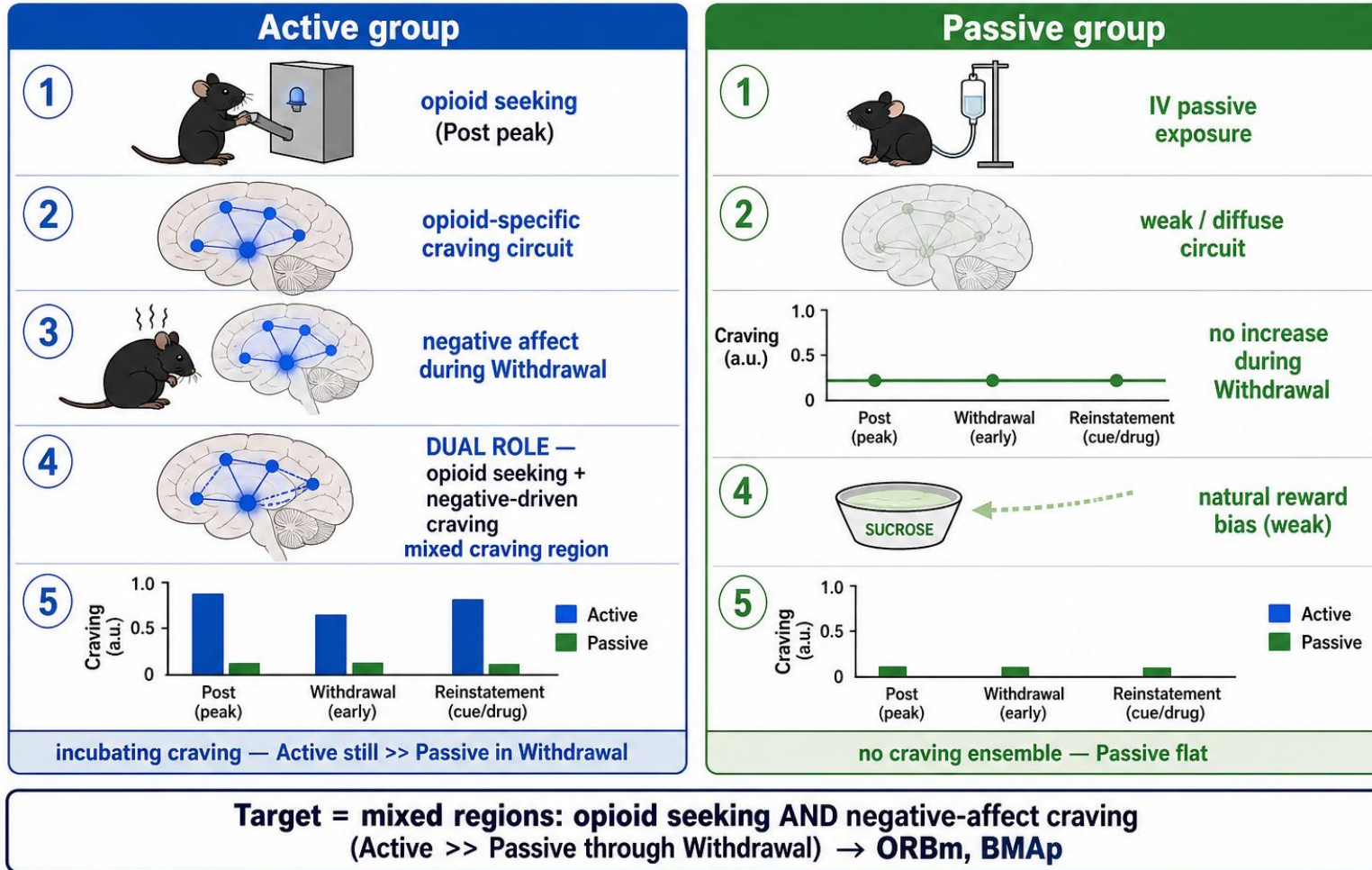
9 · Part 2 — Zoom-in logic (Finding A filter)

- After the 7-region shortlist, inspect density AND cell count across phases, Active vs Passive, every mouse.
- Keep regions where Active rises at Post, dips at WD but stays >> Passive, and Passive stays flat.
- Drop regions where Passive moves with Active (shared context) or one mouse drives the mean.
- Survivors for Finding A: ORBm + BMAp only.

CODE: `trap_region_zoomin.py` → `region_zoomin/<REG>_density_count_by_phase.png`
→ `region_zoomin/<REG>_withinphase_ActiveVsPassive.png`

10 · Finding A schematic — dual-role craving

Finding A — Why Active >> Passive even during Withdrawal?



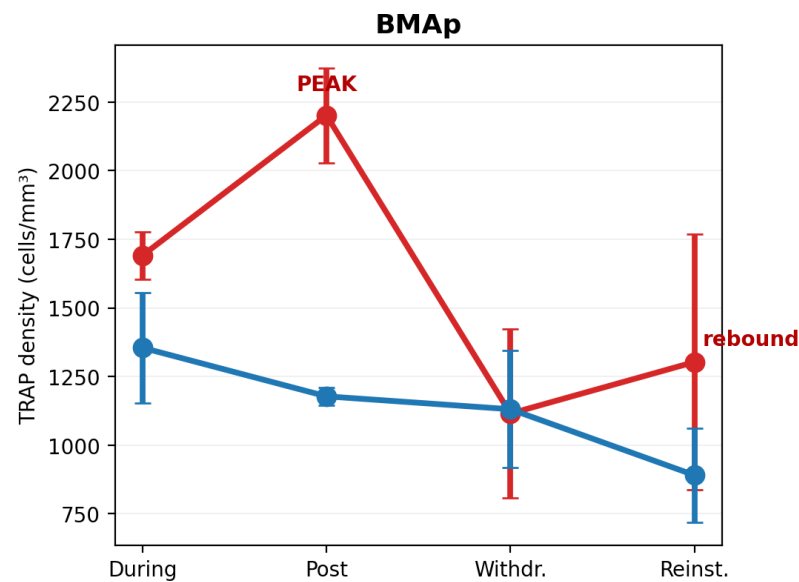
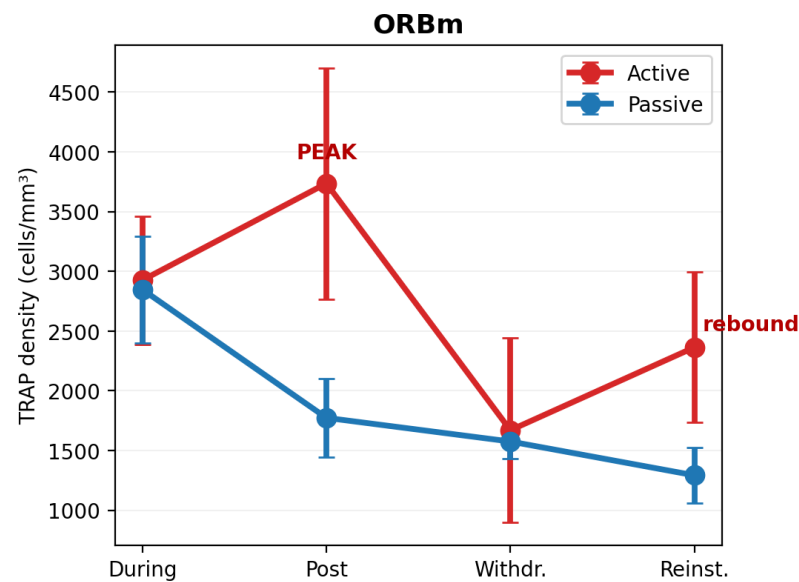
- WHAT: conceptual dual-mode craving at ORBm/BMap.
- Post: opioid seeking (Active high). WD: negative-affect craving persists while Passive flat.
- WHY shown: frames why Active>>Passive through WD is incubation-compatible.

Figure: forMark/schematic_findingA_dual_role_incubation.png

Related: trap_findingA_bridge_fig.py · trap_forMark_figs.py

11 · Finding A evidence — ORBm & BMAp raw density

Active-specific: Active > Passive, peaks at Post, rebounds at Reinstatement — Passive stays flat

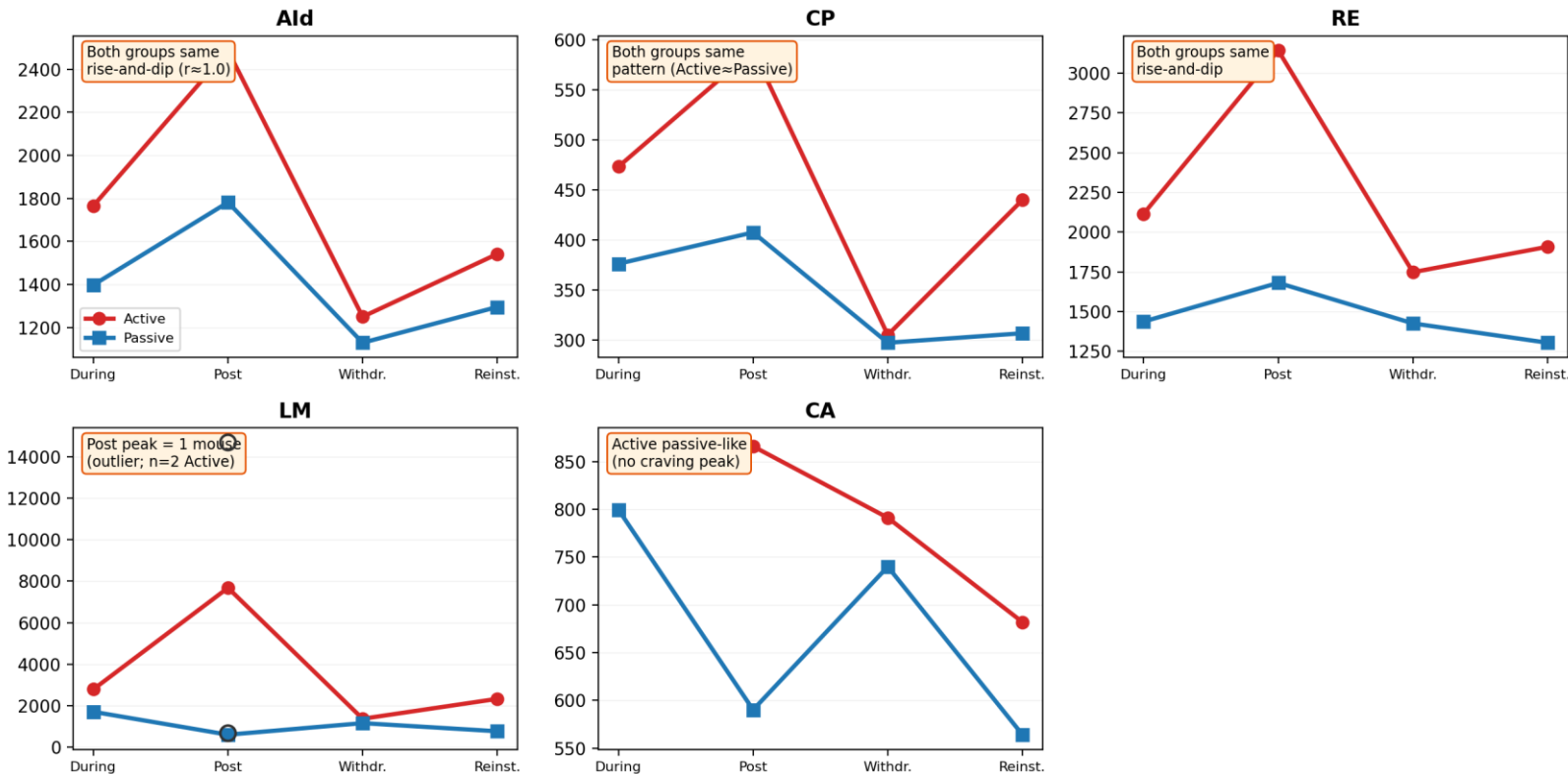


- WHAT: Active vs Passive density trajectories for ORBm & BMAp.
- Pattern: Active peaks Post, dips WD, rebounds Rein; Passive essentially flat.
- This is the Active-selective signature used to argue opioid-related seeking.

CODE: `trap_forMark_figs.py / trap_hilo_ORBm_BMAp_COAa.py`
Out: `forMark/evidence_active_ORBm_BMAp.png`

12 · Why not the other original-7 regions?

Deprioritized from original 7 — why they do NOT isolate Active craving



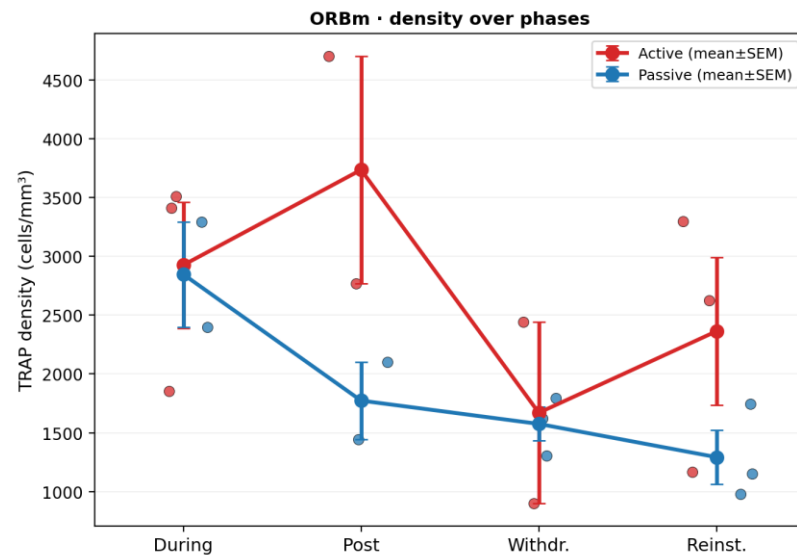
Raw group-mean density (cells/mm³) · terminal design · LM circles = individual Active Post mice (n=2)

- WHAT: summary of why CA, Ald, LM, RE, CP were deprioritized for Finding A.
- Typical failure modes: Passive not flat, both groups move together, or unstable mouse leverage.
- AI check: agent should not treat all 7 as equal final targets.

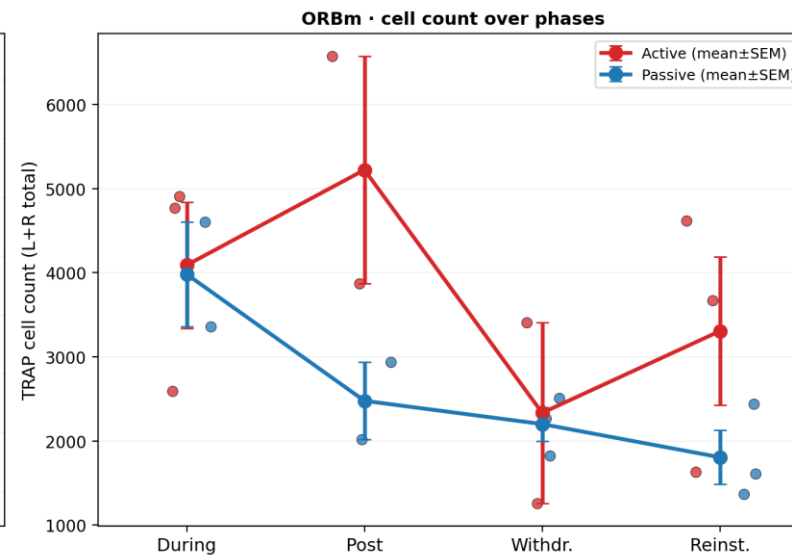
[CODE](#) / out: [forMark/deprioritized_5regions_summary.png](#) (forMark helpers)

13 · ORBm zoom-in — density & cell count

ORBm (Cluster 1) — individual mice (dots) + group mean±SEM
terminal design: each mouse = one phase; lines connect group means across (different) mice



dots = individual mice (Active red / Passive blue) · no per-mouse cross-phase lines (TRAP is a terminal snapshot, not longitudinal)



- WHAT: ORBm group-level density and cell count by phase (Active vs Passive).
- Confirms the Active-selective rise / WD dip / Rein rebound pattern.
- Density metric = calculated_mm3 (cells / sample volume).

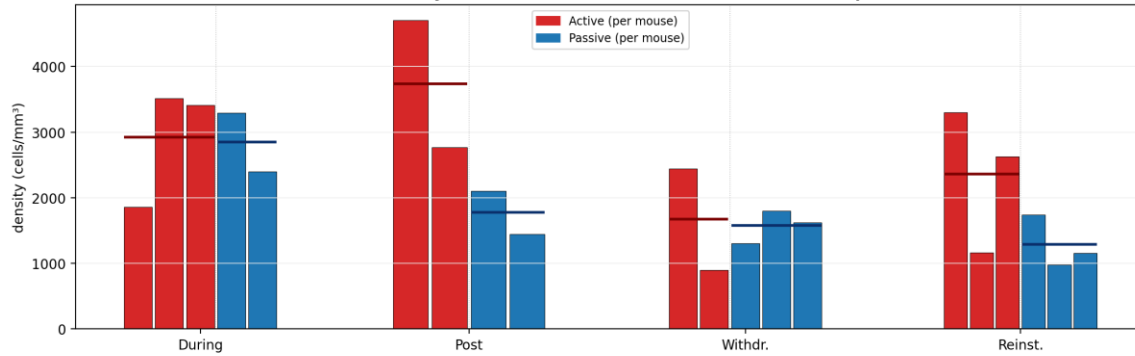
CODE: `trap_region_zoomin.py`

Out: `region_zoomin/ORBm_density_count_by_phase.png`

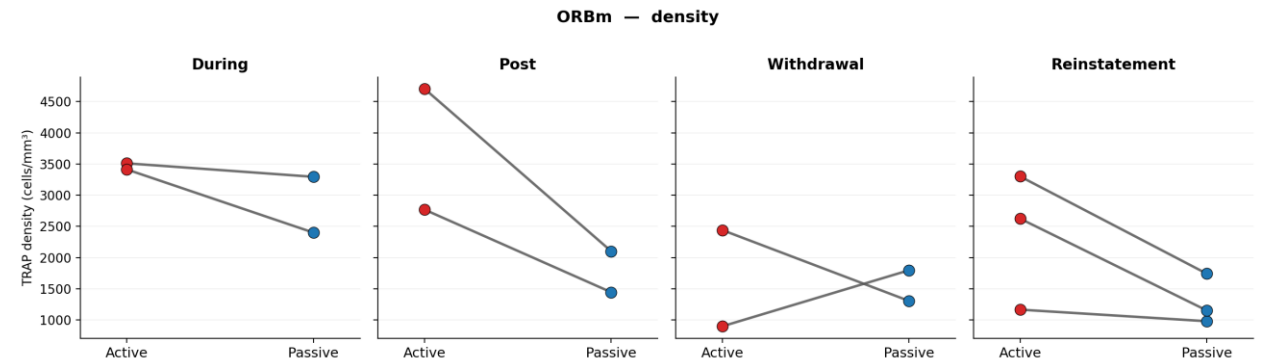
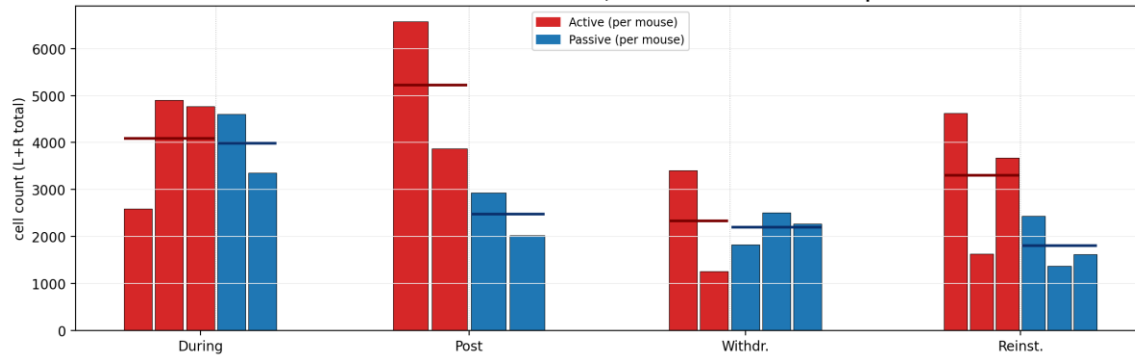
14 · ORBm mouse-level + paired slope

ORBm (Cluster 1) — within-phase Active vs Passive (one bar = one mouse)
horizontal line = group mean · comparison is WITHIN each phase (fair, same-phase mice)

ORBm · density — individual mice, Active vs Passive within each phase



ORBm · cell count — individual mice, Active vs Passive within each phase

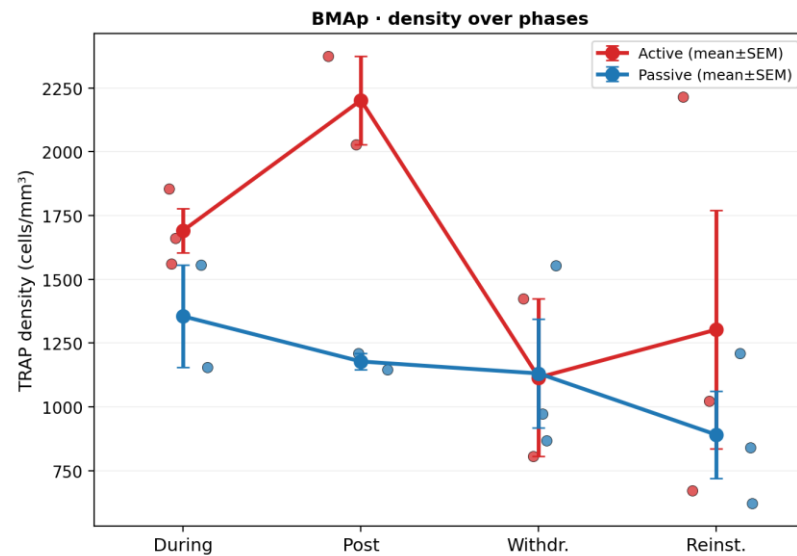


- Left: one bar = one mouse within-phase Active vs Passive. Right: paired-slope consistency across mice.
- Shows the effect is not a single-mouse artifact.

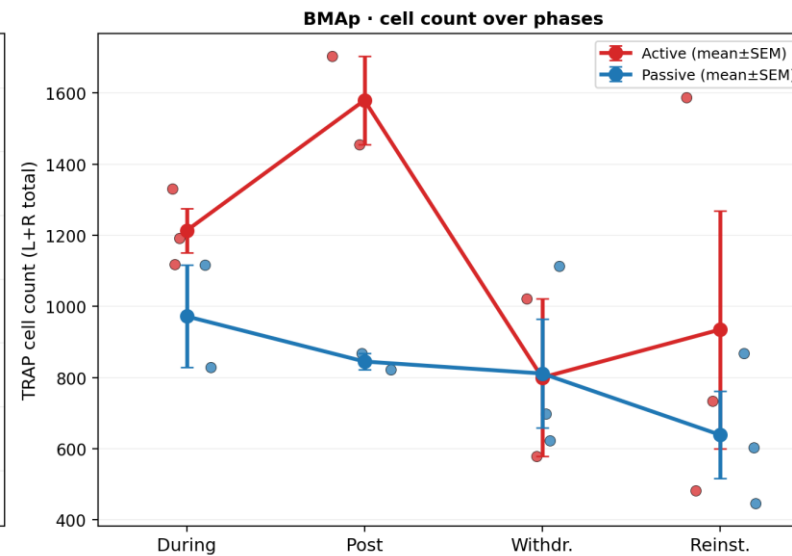
CODE: [trap_region_zoomin.py](#) · [forMark/paired_slope/ORBm_density_paired_slope.png](#)

15 · BMAp zoom-in — density & cell count

BMAp (Cluster 4) — individual mice (dots) + group mean±SEM
terminal design: each mouse = one phase; lines connect group means across (different) mice



dots = individual mice (Active red / Passive blue) · no per-mouse cross-phase lines (TRAP is a terminal snapshot, not longitudinal)



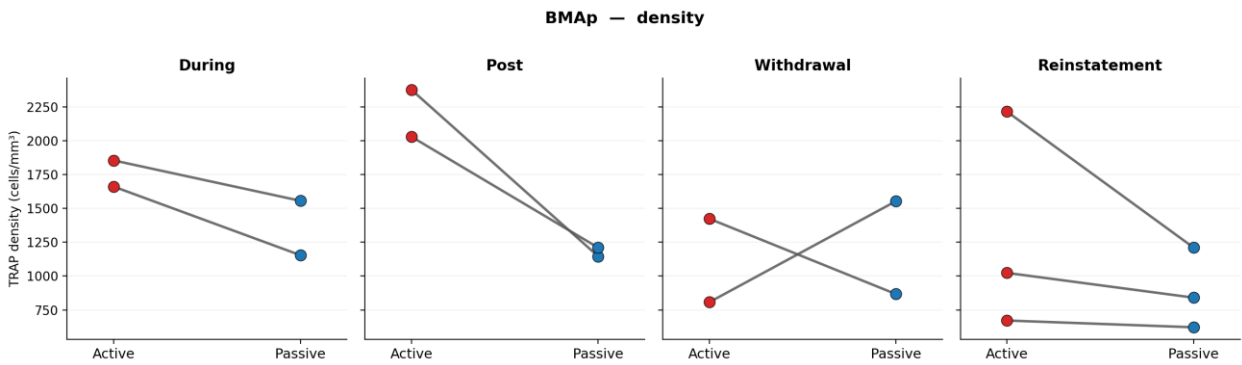
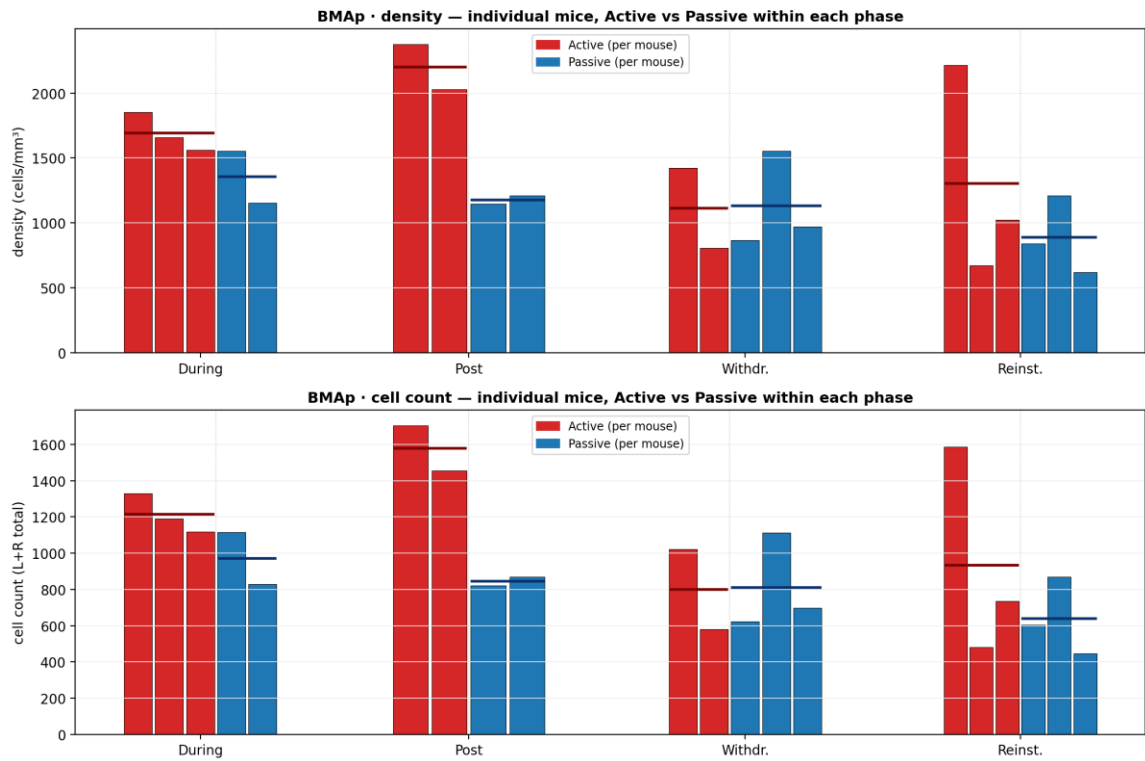
- WHAT: BMAp group-level density and cell count by phase.
- Same Active-selective / Passive-flat motif as ORBm → second Finding A region.
- Anatomically: basomedial amygdala posterior (emotional valence / addiction motivation).

CODE: `trap_region_zoomin.py`

Out: `region_zoomin/BMAp_density_count_by_phase.png`

16 · BMAP mouse-level + paired slope

BMAP (Cluster 4) — within-phase Active vs Passive (one bar = one mouse)
horizontal line = group mean · comparison is WITHIN each phase (fair, same-phase mice)



- Mouse-level Active vs Passive bars + paired slopes confirm BMAP consistency across animals.

CODE: [trap_region_zoomin.py](#) · [forMark/paired_slope/BMAP_density_paired_slope.png](#)

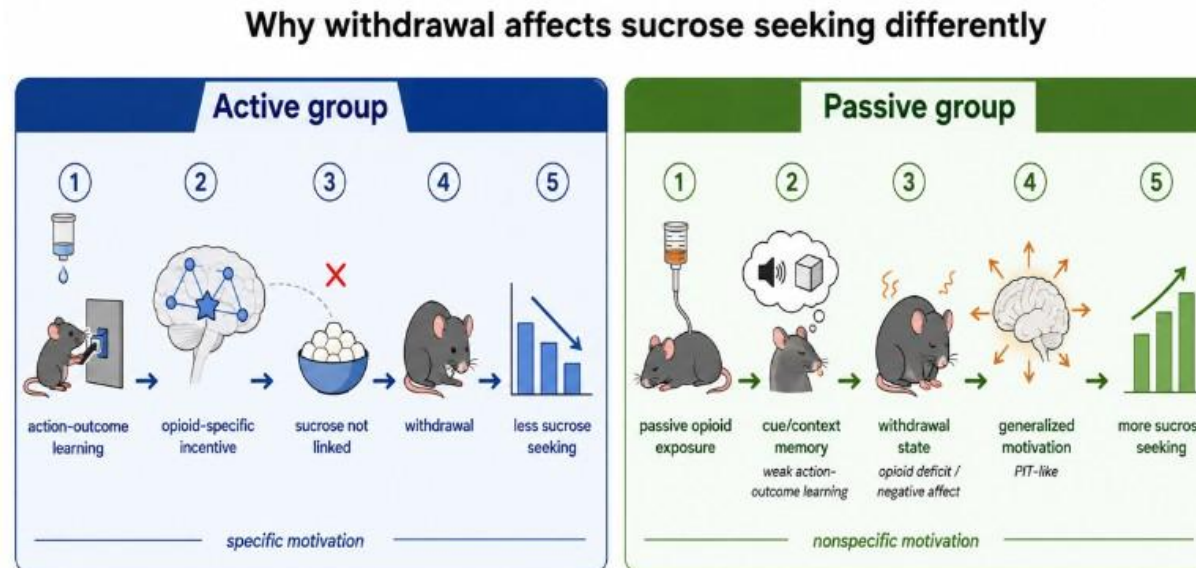
17 · Finding B start — why Passive matters

- Passive is not only a control: noncontingent exposure still engages cue/context and WD state.
- Regions with Passive > Active at Post + Withdrawal are candidates for generalized (non-opioid-selective) motivation.
- Manipulation prediction: COAa OFF may affect broad seeking (e.g. sucrose); ORBm/BMAp OFF should spare that more.
- This is a testable dissociation, not a proven causal result from TRAP alone.

18 · Finding B schematic — Passive withdrawal / patient-relevant

relevant pathway for patients, who are often first exposed to opioids noncontingently through medical treatment. Therefore, our dataset suggests that the passive group, not only the active group, may be highly informative—especially if we **identify brain regions that are selectively elevated during withdrawal only in the passive group**

- WHAT: schematic for Passive-biased / patient-relevant motif.
- Links Post+WD Passive elevation to cue/context + withdrawal state.
- Sets up cluster-3 screen for Finding B candidates.

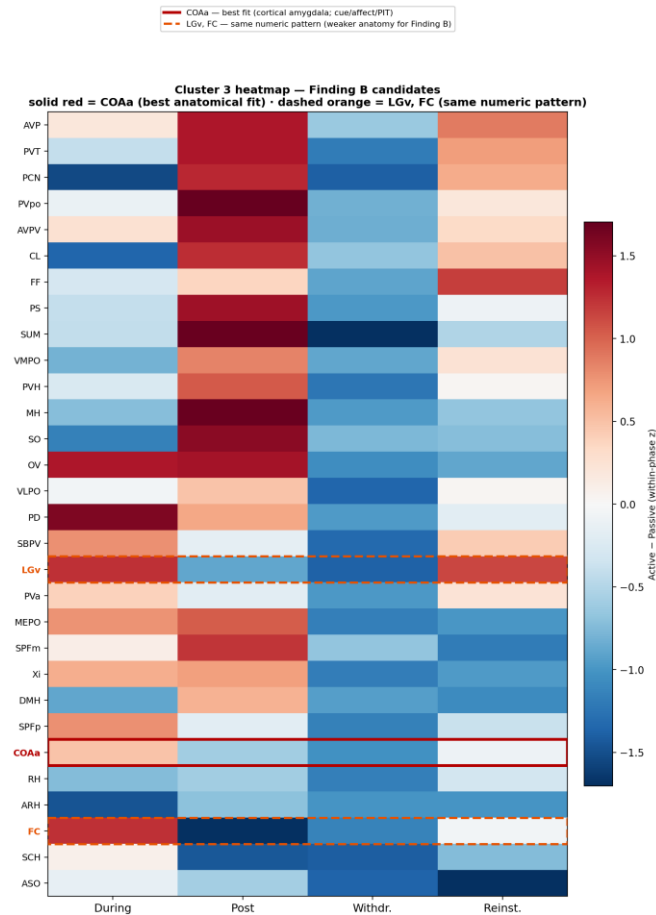


Active = opioid-specific motivation; Passive = withdrawal-driven generalized motivation.

Figure: [forMark/schematic_findingB_passive_withdrawal.png](#)

Related: [trap_findingB_cluster_map.py](#)

19 · Cluster 3 heatmap — Finding B candidates



Finding B rule: Passive > Active at Post & Withdrawal; Active > Passive at During & Reinstatement.
COAa chosen for Finding B schematic (cue/context at Post + withdrawal negative affect → generalized motivation candidate).
Cluster 3 was not selected for Finding A (Active craving filter) — separate screen.

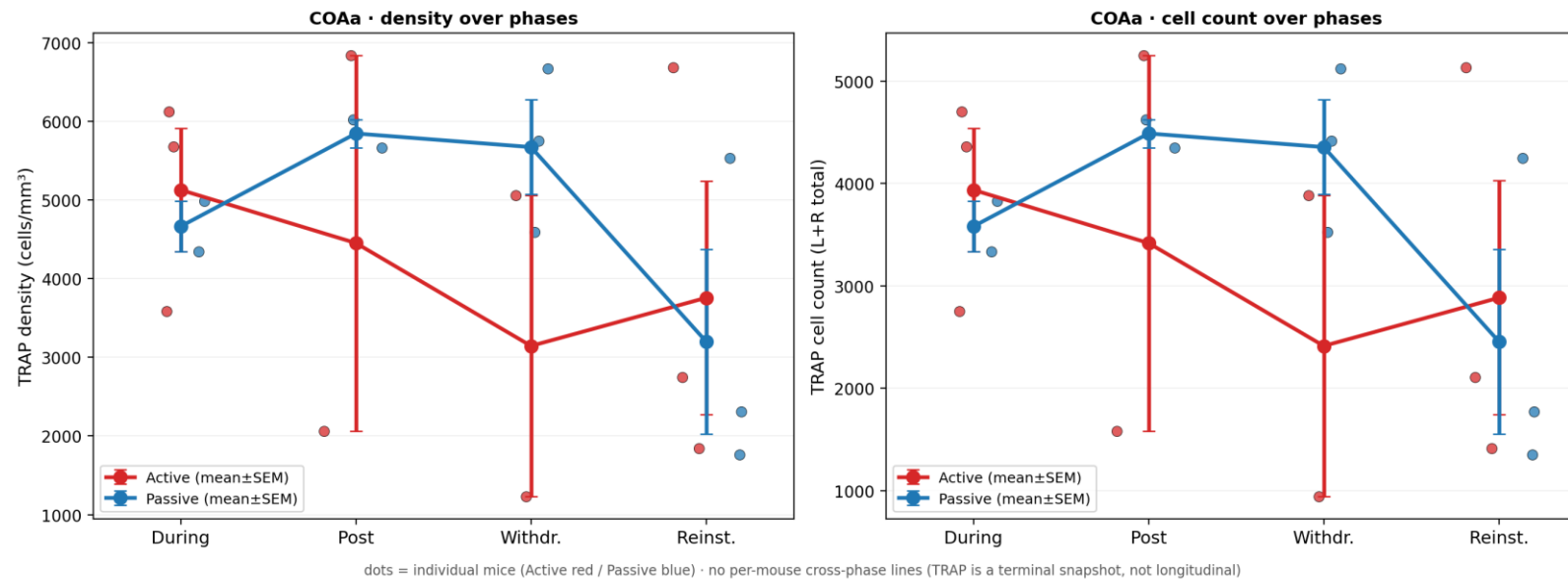
- WHAT: cluster-3 region heatmap / ranking for Passive-biased pattern.
- COAa chosen as best anatomical fit (LGv/FC can match numerically).
- AI check: agent must produce a second Passive-biased story, not only Finding A.

CODE: `trap_fig_cluster3_heatmap.py · findingB_COAa_revision/`
Out: `Fig_cluster3_FindingB_heatmap.png`

20 · COAa zoom-in — density & cell count

- WHAT: COAa density and cell count by phase, Active vs Passive.
- Pattern: Passive > Active especially at Post and Withdrawal.
- This is Finding B's primary quantitative panel.

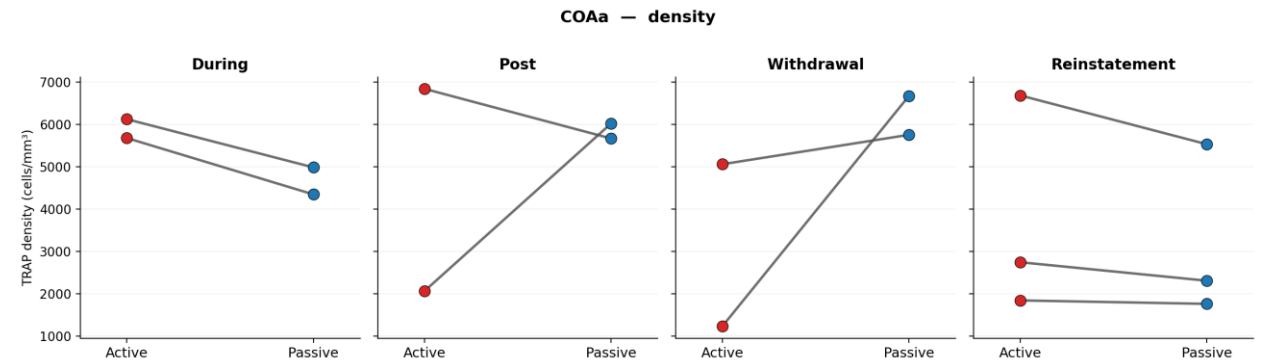
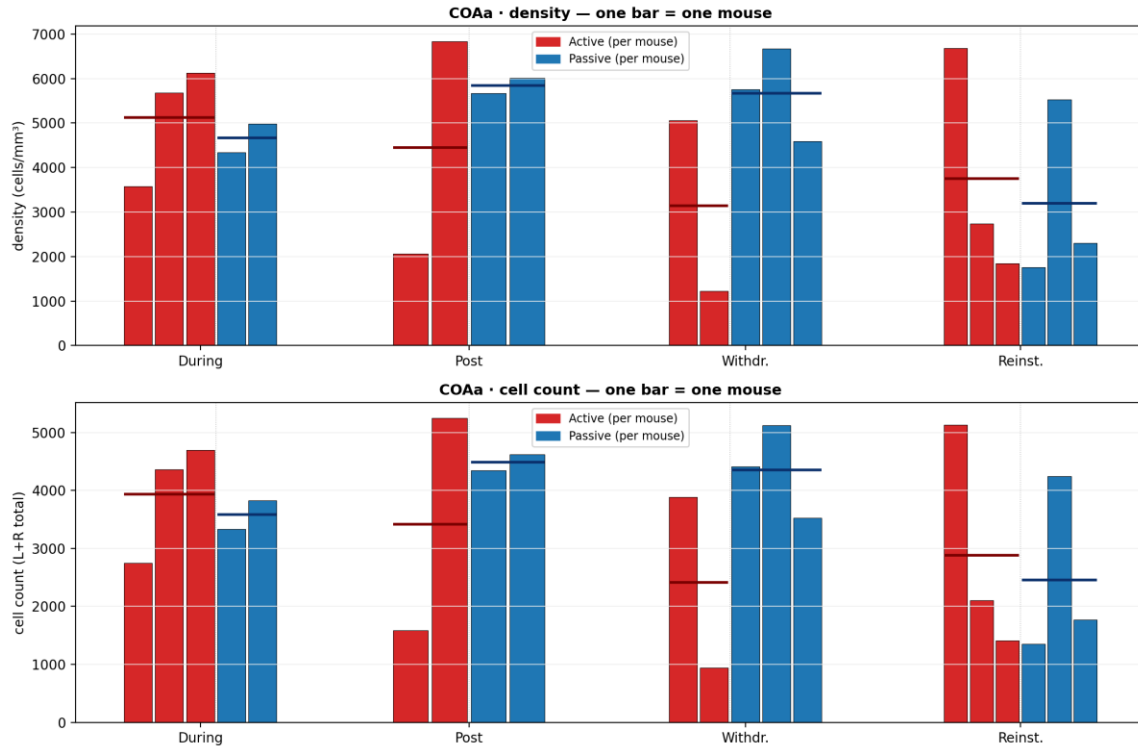
COAa (Cluster 3) — individual mice (dots) + group mean $\pm$ SEM
terminal design: each mouse = one phase; lines connect group means across (different) mice



CODE: COAa family / P>A helpers (see `trap_build_pptx_COAa_family.py`)
Out: `forMark/COAa_family/COAa_density_count_by_phase.png`

21 · COAa mouse-level + paired slope

COAa (Cluster 3) — within-phase Active vs Passive
horizontal line = group mean · orange interest: Passive > Active at Post & Withdrawal



- Mouse-level and paired-slope panels for COAa Passive > Active consistency.

CODE: [P_gt_A_PostWD helpers · forMark/paired_slope/COAa_density_paired_slope.png](#)

22 · Code map (where to click on GitHub)

- Pipeline drivers: `init_TRAP_pipeline.m` · `trap_config.m` · `RUN_PIPELINE_ALL.m`
- Clustering: `Step_03_region_clustering_PCA_kmeans/` · `Step_13_universal_cluster_viz/`
- Final-story Python (repo root): `trap_region_selection_ppt.py` · `trap_region_zoomin.py` · `trap_build_pptx.py`
- Finding helpers: `trap_fig_cluster3_heatmap.py` · `trap_hilo_ORBm_BMAp_COAa.py` · `trap_forMark_figs.py`
- Pack offline copies: `colleague_AI_benchmark_pack/code/`
- Canonical outputs: `TRAP_OUTPUT_calculated_mm3/region_selection_PPT/` · `region_zoomin/` · `forMark/`

23 · AI-benchmark rubric (what success looks like)

- Recover biological aim: Active>>Passive primary + Passive-biased secondary.
- Reproduce funnel: forebrain → K=4 → keep clusters 1&4 → shortlist 7 → zoom survivors ORBm/BMAp.
- Name Finding A = ORBm + BMAp with Passive-flat Active-selective rationale.
- Name Finding B = COAa (Passive > Active at Post+WD); mention LGv/FC as numeric peers.
- Point to the correct scripts above for each figure — not the entire Step-13 PNG dump.
- Do not treat histology slides as pipeline-generated plots (they were inserted into the PPT).

24 · Summary

- Aim: map opioid-selective craving circuits vs generalized / Passive-side motivation.
- Method: cluster first (structure) → shortlist → zoom-in (individual regions + mice).
- Finding A: ORBm & BMAp — Active >> Passive, Passive flat.
- Finding B: COAa — Passive > Active at Post + Withdrawal.
- This deck = figure + meaning + code for the final story only.
- Companion docs in pack: 00_START_HERE.md · HOW_FINAL_PDF_WAS_MADE.md · FINAL_PDF_FIGURE_MAP.md